

EXAM

Video Processing
Course number 0512-4263-01
Spring 2022
Faculty of Engineering
Tel-Aviv University
Prof. Shai Avidan

DATE: 1/8/2022
TIME: 9am
DURATION: 3 hours

Read all of the following information before starting the exam:

- Show all work, clearly and in order.
- Justify your answers.
- Please keep your written answers brief; be clear and to the point.
- Each answer should take one page only (text, images and equations).
- This exam has 8 questions and is worth 100 points.
- Good luck!

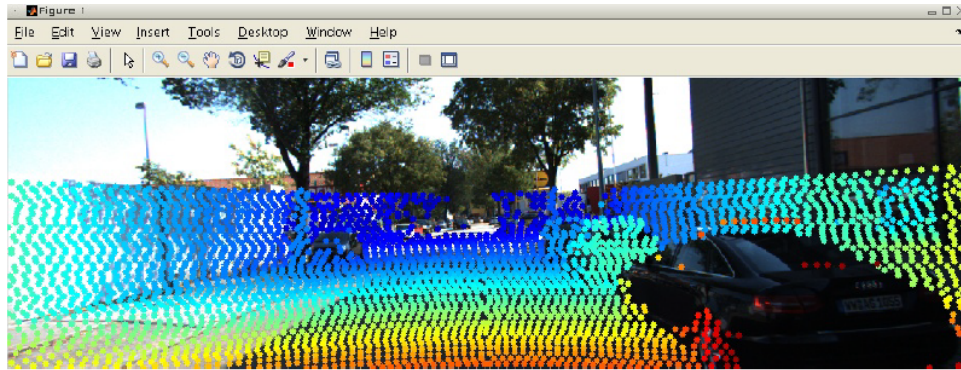
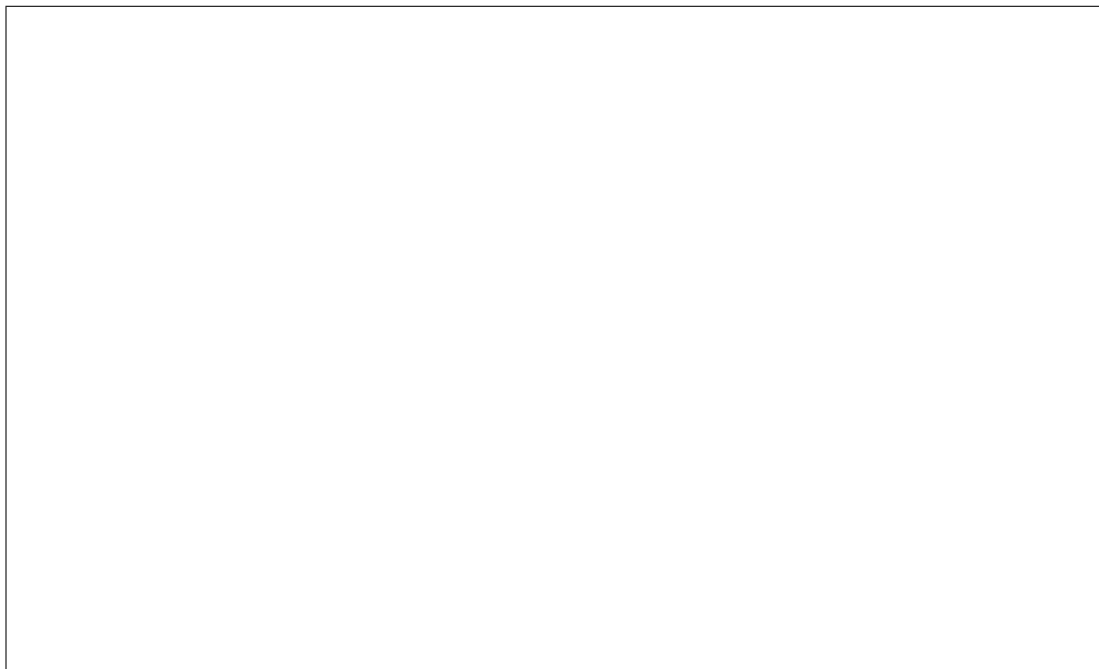


Figure 1: An image with LiDAR measurements superimposed on it.

1. (13 points) OPTIC FLOW

A vehicle is equipped with a grayscale camera and a LiDAR sensor. The LiDAR sensor provides depth value to a *sparse* subset of the pixels in the image (i.e., not every pixel has a corresponding depth value). See example in Figure 1.

Modify the Horn-Schunck algorithm to take as input two grayscale images I_1 and I_2 and their corresponding sparse depth maps D_1 and D_2 , and produce the optic flow between them.



2. (12 points) TRACKING

Write down the Markov model assumption and the Conditional Independence of observations assumption and show that:

$$p(\mathbf{x}_{1:t}|\mathbf{z}_{1:t}) \propto p(\mathbf{z}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{x}_{t-1})p(\mathbf{x}_{1:t-1}|\mathbf{z}_{1:t-1}) \quad (1)$$



3. (13 points) DYNAMIC TEXTURE

Word2Vec is a neural network that maps English words to vectors in some Euclidean space. Use *Word2Vec* to synthesize text. That is, your algorithm should take as input a short text (i.e., a sentence or a paragraph) and generate as output a long text (i.e., a page of text).



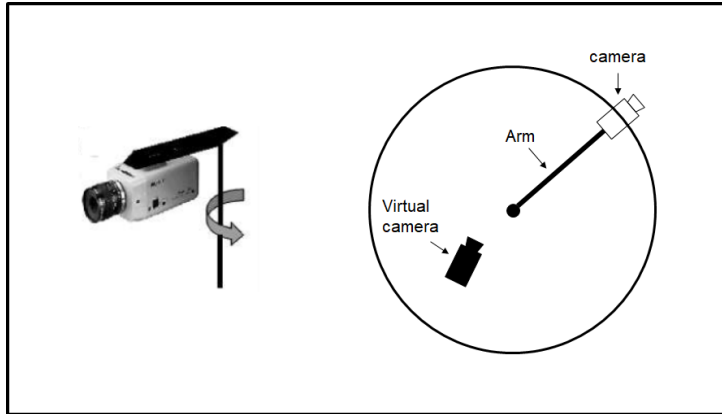


Figure 2: (Left) A camera on an arm is recording a video as it rotates around an axis. (Right) top view of the set up.

4. (12 points) LIGHT FIELDS

Consider Figure 2. On the left you see a camera attached to an arm that is rotating around an axis. The camera is recording a video sequence as it is rotating. On the right, you see a schematic top view of the setup. Suggest a way to synthesize a novel view from the virtual camera inside the circle, using the images taken by the rotating camera.



5. (13 points) PATCHMATCH

PatchMatch returns, for every patch in the source image, its nearest neighbor in the target image. We can term this as 1-NN PatchMatch.

suggest how to extend it to k -NN PatchMatch that returns, for each patch in the source image, a list of the top k Nearest Neighbors in the target image (k is a parameter supplied by the user).

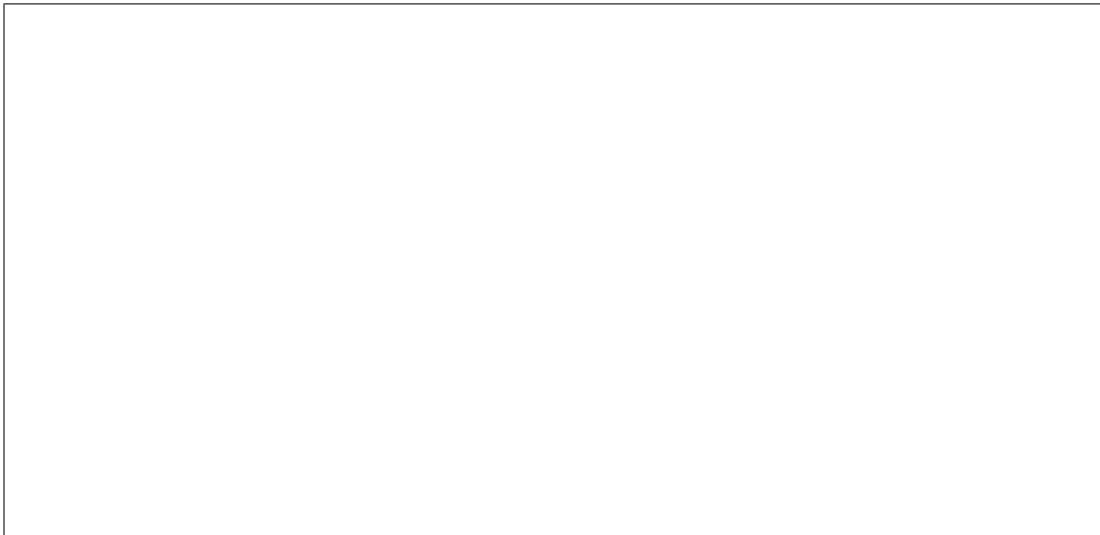




Figure 3: Image of Lena. The user clicks on points **A** and **B**. The system automatically draws the black line between them that goes along the edge of the hat.

6. (12 points) GEODESIC DISTANCE

Consider Figure 3. The user clicks on points *A* and *B* and the system automatically draws the black path between them that follows the edge of Lena's hat. Suggest a way to compute the path between these two points, as shown in the figure.



7. (13 points) SEGMENTATION

Consider a Social network defined as a graph $G = (V, E)$, where nodes $v \in V$ denote users and edges $(u, v) \in E$ denote connections between users. The weight function $w(u, v) : E \rightarrow \mathbb{R}$ defines the strength of the connection between users u and v .

Assume the owner of the graph marks a few of the users (i.e., nodes) with a label (say, user u_1 is a financial expert and user u_2 is a fashion expert). We wish to learn, for each node in the graph his/her mix of interests (i.e., user v is 30% interested in finance and 70% interested in fashion). Suggest an algorithm for doing that.

8. (12 points) CLUSTERING

Is the k-means algorithm guaranteed to terminate? explain your answer.

EMERGENCY BOXES

A large, empty rectangular box with a thin black border, intended for providing emergency contact information.A second large, empty rectangular box with a thin black border, identical to the one above, for additional emergency information.